

Intelligent Disease Surveillance System

User Manual — how to use the dashboard

Live app: disease-surveillance-early-outbreak-detection.streamlit.app

1. Getting started

The system runs in any web browser — there is nothing to install. Just open the link:

disease-surveillance-early-outbreak-detection.streamlit.app

- **Six tabs** run across the top of the page. Click a tab to switch views.
- Just under the title, three coloured dots show which data services are live — **SQLite** (always on), **MongoDB**, and **Redis**. Green = connected.
- If a page ever looks stuck, refresh the browser tab.

2. The six tabs at a glance

Tab	What it is for
Overview	The summary screen — total cases, confirmed cases, and how well the model performs.
Outbreak monitor	Which states are at risk right now, with a one-click alert to health officials.
Case Registration & Triage	Enter a new suspected case and get an instant risk assessment, then save it.
Trends	Charts of cases, deaths, risk and alerts over time — filterable.
Notifications	The log of alerts sent to the people in charge, with an acknowledge button.
Model Status	Technical details of the adaptive AI model and its accuracy.

3. Registering a suspected case (the main task)

This is the heart of the system. Open the Case Registration & Triage tab and follow these steps:

1. **Fill in the basic case information** (always visible): tick the patient's symptoms, then choose age group, sex, suspected disease, state, LGA and the report date.
2. **Add more detail if you have it** by clicking to expand any of these sections:
 - **Environmental factors** — temperature, rainfall, flooding, rodent activity, travel, contacts.
 - **Clinical notes** — free-text notes from the clinician.
 - **Laboratory information** — sample ID, PCR result, technician.

- **Supporting documents** — upload lab reports or images (PDF, DOCX, PNG, JPG).
3. Click **Predict**. The model shows the Lassa likelihood, a risk level, a confidence, and a recommended action.
 4. Click **Save case**. The record is stored and you get a confirmation with a unique case ID.

Behind the scenes the structured details go to **SQLite**, the notes and documents go to **MongoDB**, and a new-case event is published to **Redis** — all linked by the same case ID.

4. Reading the Outbreak monitor

5. Choose a disease at the top (Lassa fever has the full history).
6. The table ranks each state with a **Risk level** (● HIGH · ● MEDIUM · ● LOW), a **Confidence**, a **Trend** arrow (↑ rising, ↓ falling, → steady), and a **Recommended action** (Monitor / Investigate / Notify NCDC).
7. Click “**Notify NCDC ...**” to log an alert for every HIGH-risk state. The alerts appear on the Notifications tab.

5. Trends

Use the filters (disease, state, year) and the **Show** selector to switch the chart between **Confirmed cases**, **Deaths**, **Risk score**, and **Alerts**. Pick a single state to see the model's risk curve over time.

6. Notifications

This is the alert log sent to the people in charge (State Epidemiologist, NCDC Duty Officer). Each row shows the time, location, severity, recipient, delivery method and status. To mark an alert as handled, pick its number under **Acknowledge alert #** and click **Mark acknowledged**.

7. Model Status

For the technically curious: the three adaptive SVMs and their accuracy. The headline is that the **adaptive (always-learning) model matches a traditional train-once model** — outbreak prediction about **89%** — so continuous learning costs almost no accuracy.

8. What the coloured dots mean

Dot	Meaning
 Green	Connected and in use.
 Grey	Not connected. The app still works on SQLite; for MongoDB/Redis it shows what it would have stored or sent, so nothing breaks.